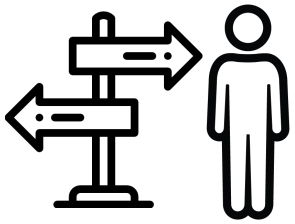
MGMT 3130
Judgment and Decision Making in Organizations

Unit 2.5
Prospect Theory



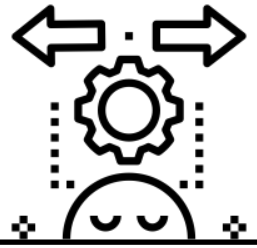
Where are we now?

Part 1: Decision Analysis

- Unit 1.1 Defining your decision problems
- Unit 1.2 Decisions involving multiple objectives
- Unit 1.3 Decision making under uncertainty

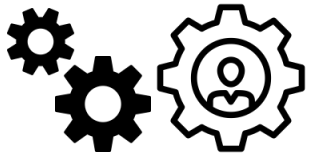
Part 2: Common Traps of Human Irrationality

- Unit 2.1 Two systems of thinking
- Unit 2.2 Availability heuristic
- Unit 2.3 Representativeness heuristic
- Unit 2.4 Bounded awareness



- **Unit 2.5 Prospect theory**
- Unit 2.6 Planning fallacy
- Unit 2.7 Escalation of commitment
- Unit 2.8 Temporal decision making
- Unit 2.9 Biases in group decision making

Part 3: JDM Issues in Contemporary Organizations



- Presentation day 1: Unethical behavior
- Presentation day 2: Going green
- Presentation day 3: Diversity and inclusion
- Presentation day 4: Performance appraisal

Unit 2.5 Prospect Theory

Prospect theory value function, v(x)

- Reference dependence
- Diminishing sensitivity
- Loss aversion

Prospect theory weighting function, w(p)

- Certainty effect
- Possibility effect

Using prospect theory, EU(option) = Σ v(x_n) w(p_n), to predict risk preferences.

PROSPECT THEORY

Prospect theory describes how people choose between different options (or prospects). It's a theory of the *psychology of choice*.

Our evaluation of a prospect can be stated as:

$$\mathbf{EU(option) = \Sigma\ v(x_n)\ w(p_n)}$$

where $v(x_n)$ is the subjective value of the various outcomes, as described in the prospect theory value function,

and $w(p_n)$ is the decision weight attached to the objective probability, as described in the prospect theory weighting function.

PROSPECT THEORY VALUE FUNCTION

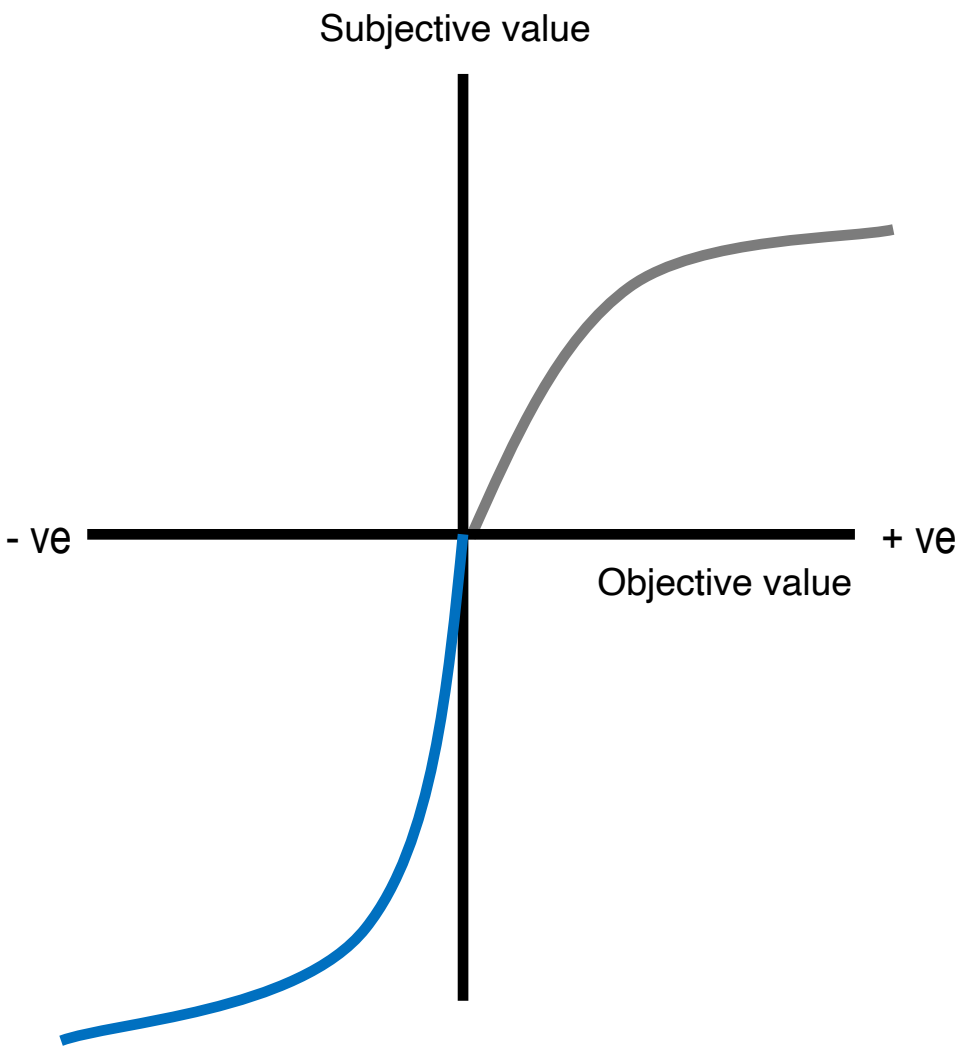
Prospect theory value function assigns *subjective values* to the *objective values* of outcomes relative to a psychologically neutral reference point.

The value function has 3 properties:

1) Reference dependence: Possible outcomes are divided into gains and losses by a reference point.

2) Loss aversion: The value function is steeper for losses than for gains.

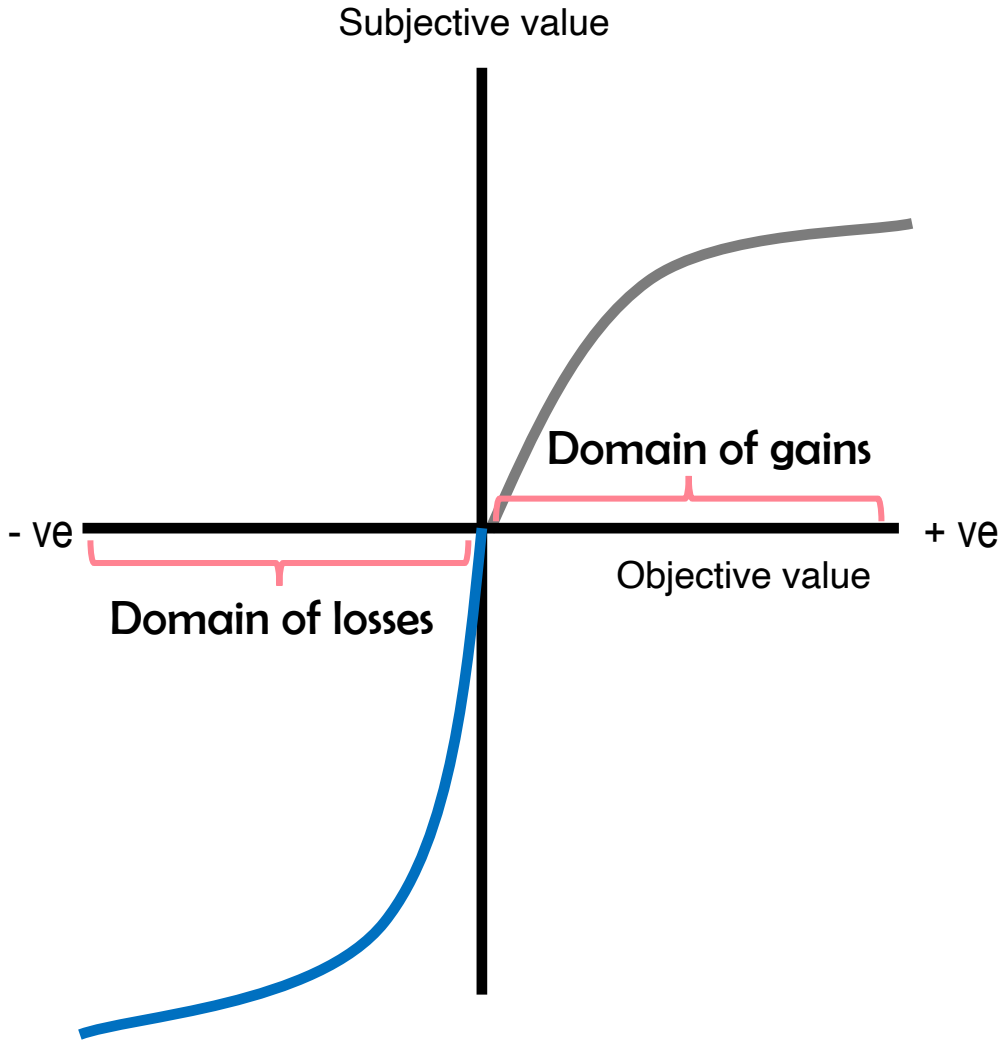
3) Diminishing sensitivity: The subjective values of both positive and negative outcomes have diminishing sensitivity.



Reference Dependence

Your evaluation of an outcome is dependent on the *reference point* that you are comparing the outcome to.

What are the potential reference points?
Whatever accessible comparison standards!

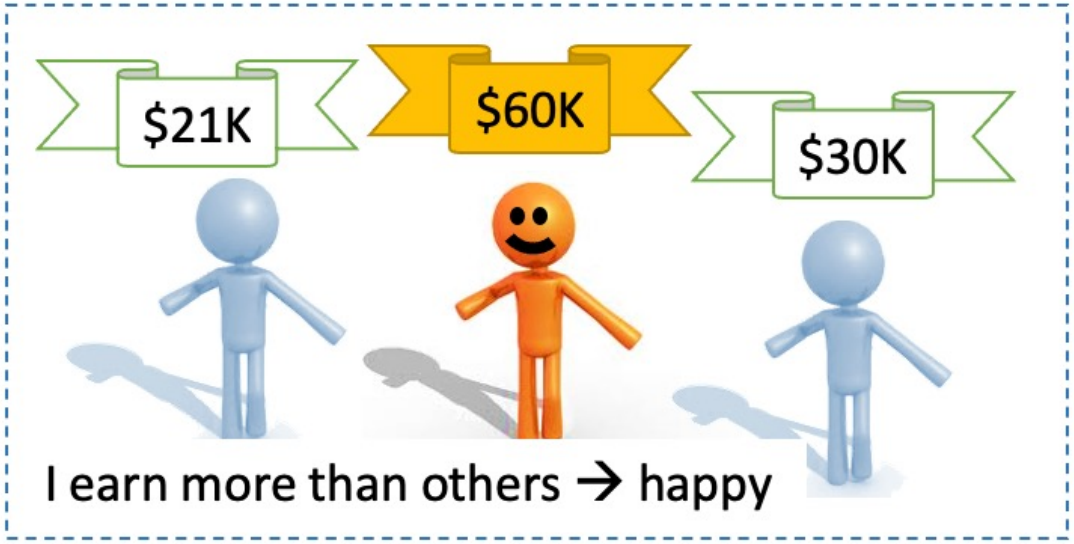


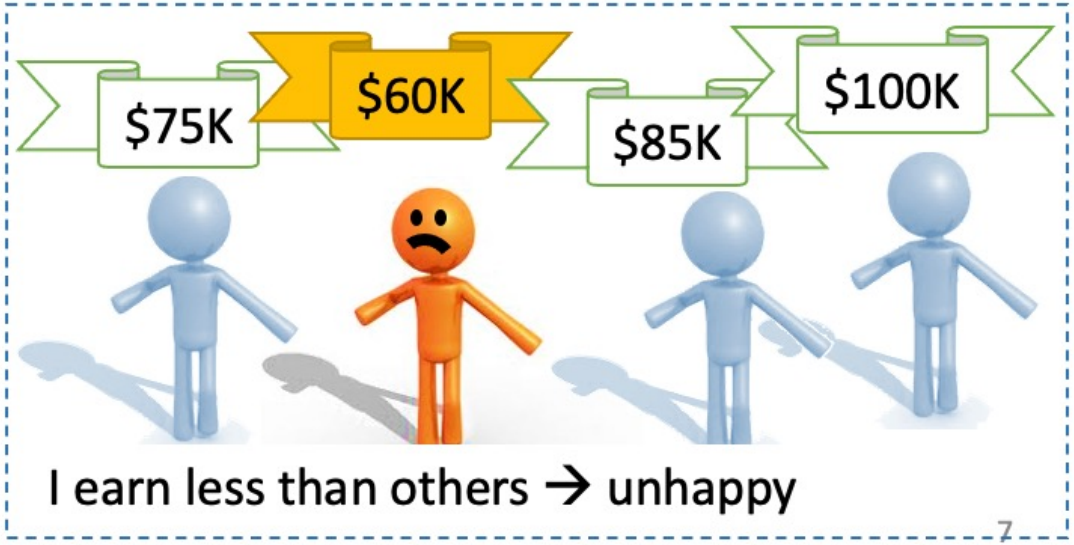
Your reference point can be a *self-generated counterfactual outcome* – what might have occurred.

- Bronze-medal winners: I might have won nothing → happy
- Silver-medal winners: I might have won the Gold-medal → unhappy

Medvec, V. H., Madey, S. F., & Gilovich, T. (1995). When less is more: Counterfactual thinking and satisfaction among Olympic medalists. *Journal of Personality and Social Psychology*, *69*(4), 603-610.

Your reference point can be the *outcome of other people*.





Your reference point can be the *status quo* (the existing/current state).

- Have a look at the following prospect: 50% chance to win $100, 50% chance to win nothing.
- If you use the status quo as a reference point: You will be happy if you end up win $100, and not extremely disappointed if you win nothing.

Your reference point can also be *the most probable outcome*.

- Consider the following prospect: 99% chance to win $100, 1% chance to win nothing.
- If you use the most probable outcome (i.e., winning $100) as a reference point: You will not be too happy if you end up win $100. And, you will be extremely disappointed if you win nothing, you feel like you *lose* $100.

Loss Aversion

Losses are disliked more than equal-sized gains are liked!

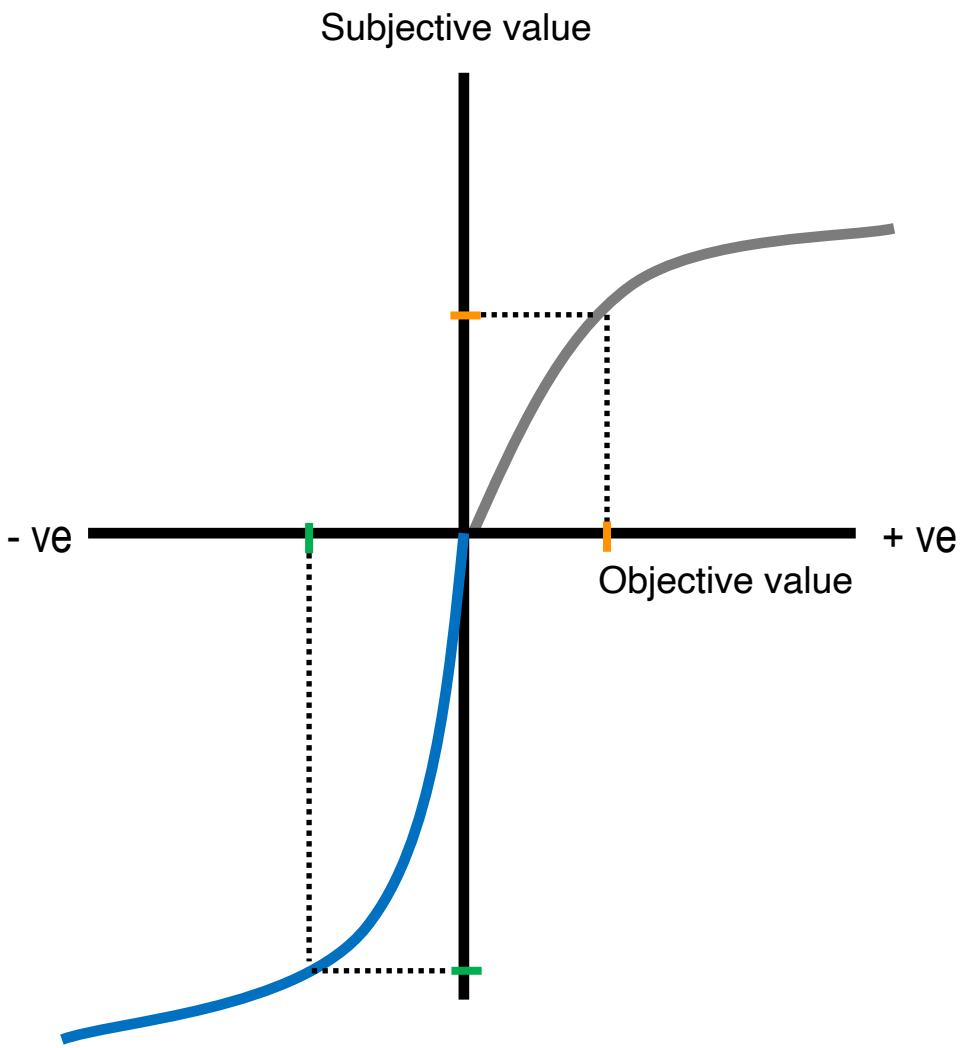
Consider a gamble:

- 50% chance to win $100,
- 50% chance to lose $100.

Will you take the gamble? No, because the *pain of losing $100* is larger than the *pleasure of winning $100.*

Loss aversion has an evolutionary history: Organisms that treat threats to be more urgent than opportunities have a better chance to survive and reproduce.

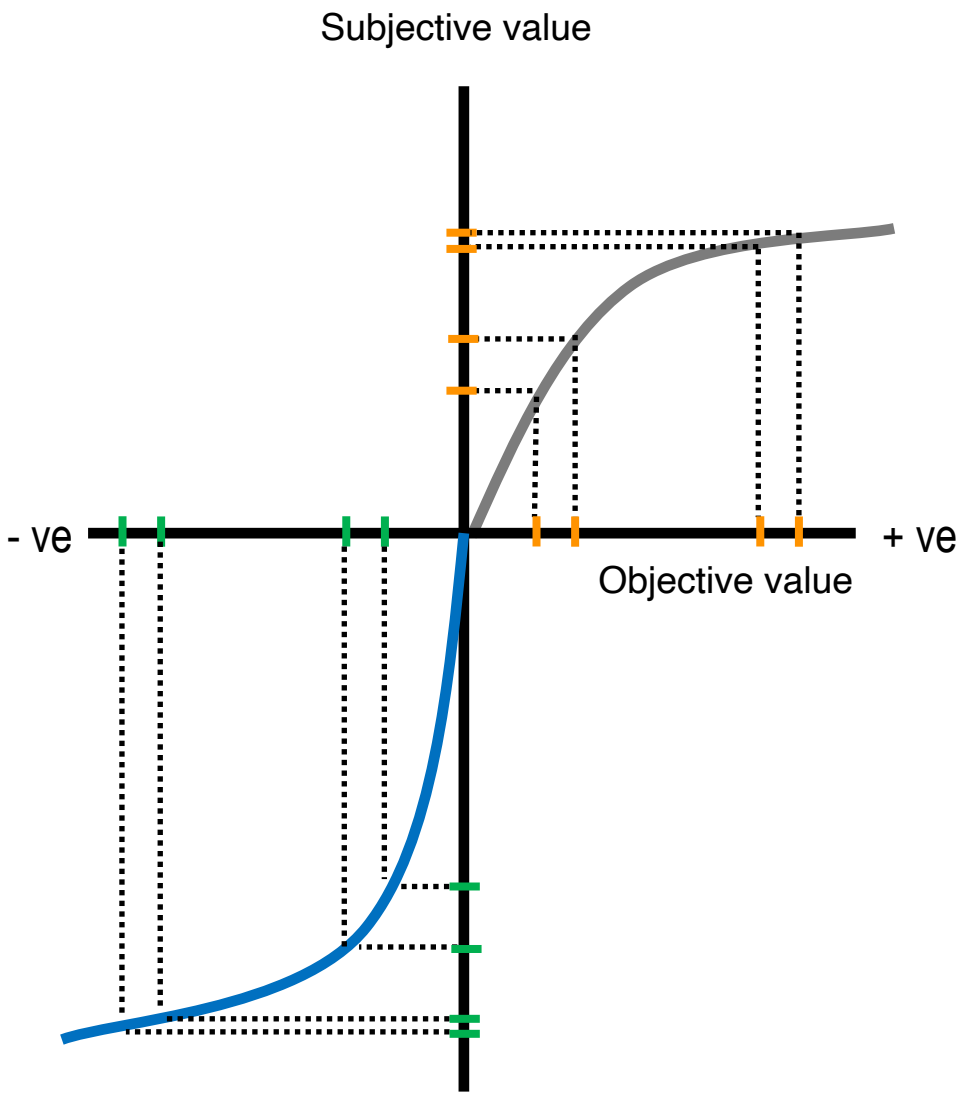
[Let's play a game to illustrate this point.]



Diminishing Sensitivity

The more units of a stimulus you are exposed to, the less you are sensitive to an additional unit.

- A hungry person may crave a steak, but having eaten one steak, a second steak will be less appealing than the first one.
- Lighting a candle has a large effect in a dark room, but the same increment of light may be undetectable in a brightly illuminated room.
- Winning (losing) an additional $1000 is a big deal if you have so far won (lost) $1000, but less of a big deal if you have so far won (lost) $100,000,000.



- Consider the following situations:

Situation 1

You need to buy a calculator.

You go to *Fortress* in *Hang Hau* and find the calculator you want for $500.

The salesperson told that you could get the same calculator for $400 if you go to the *Mongkok branch* to get it, but it will take you 1 hour by bus.

What will you do?

- Purchase the calculator at *Fortress Hang Hau* for $500, OR
- Travel to *Mongkok* and purchase the calculator for $400.

Situation 2

You need to buy a computer.

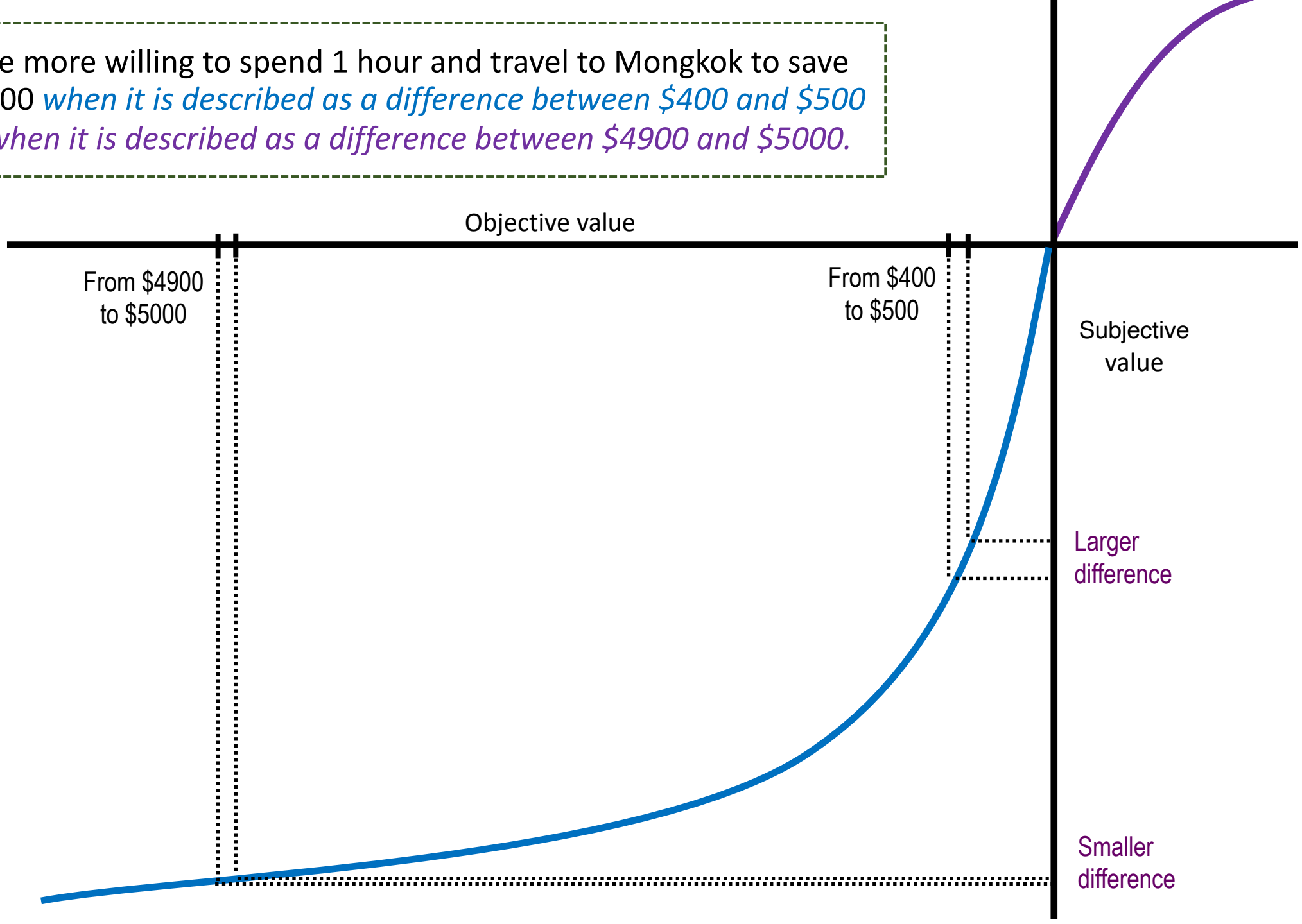
You go to *Fortress* in *Hang Hau* and find the computer you want for $5000.

The salesperson told that you could get the same computer for $4900 if you go to the *Mongkok branch* to get it, but it will take you 1 hour by bus.

What will you do?

- Purchase the computer at *Fortress Hang Hau* for $5000, OR
- Travel to *Mongkok* and purchase the computer for $4900.

We are more willing to spend 1 hour and travel to Mongkok to save that $100 *when it is described as a difference between $400 and $500* than *when it is described as a difference between $4900 and $5000.*



Phenomena Explained By The Prospect Theory Value Function

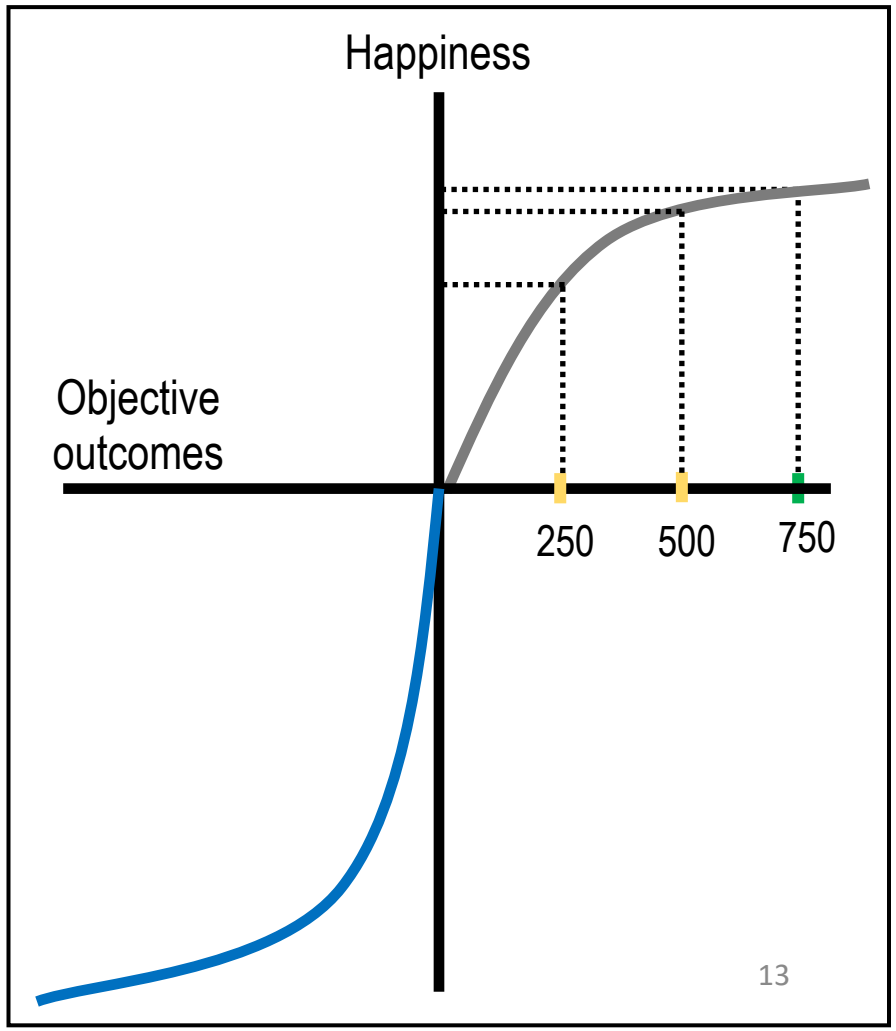
The prospect theory value function can explain a range of human irrationalities in judgment and decision making, including:

❖ *Principles of hedonic framing*

When we experience two outcomes X and Y, how should we frame them to maximize utility/happiness? *Jointly*, i.e., v(X+Y), or *separately*, i.e., v(X) + v(Y)?

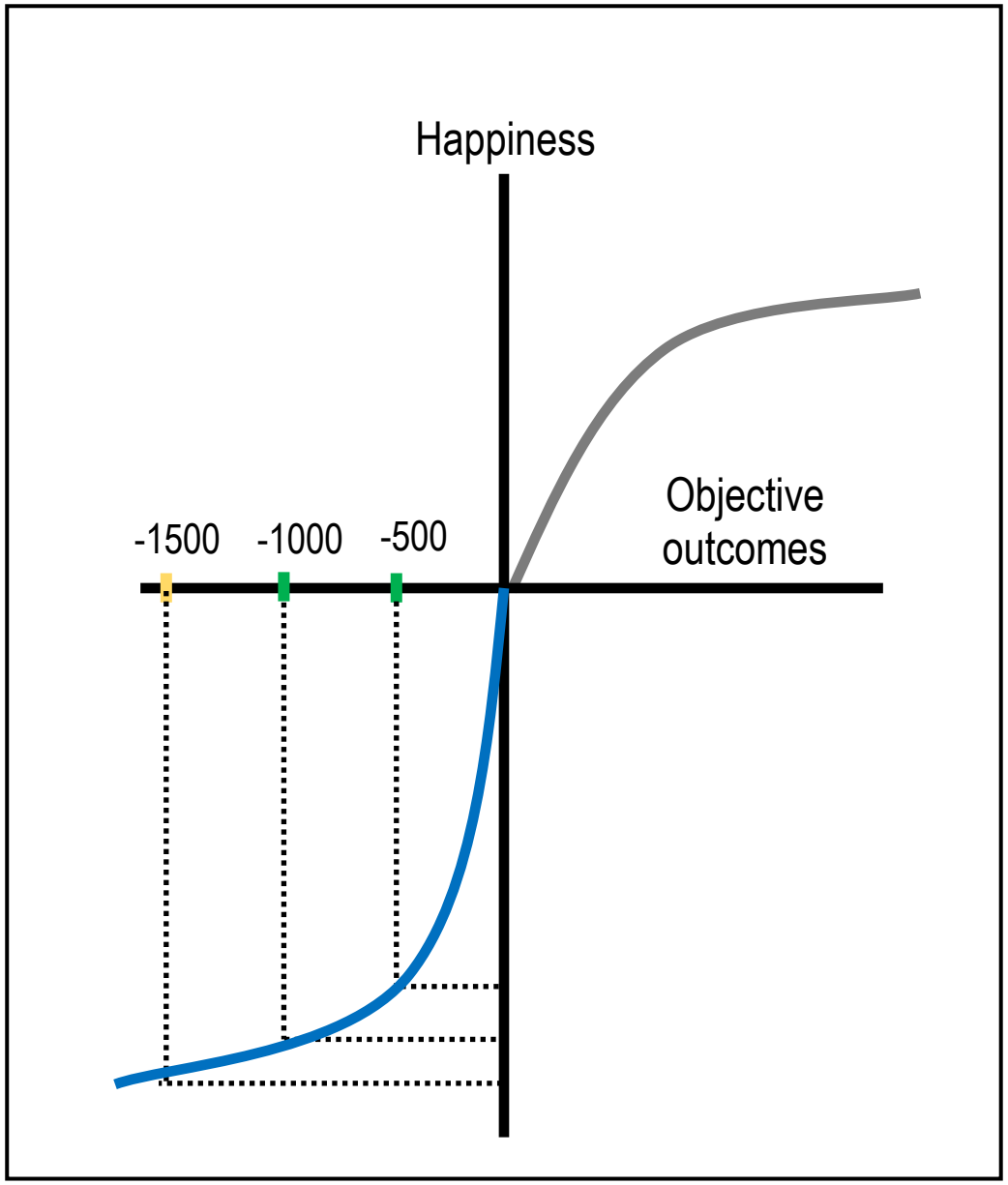
Case 1:

- *Mr. A was given 2 lottery tickets. He won $500 in one lottery and $250 in the other.*
- *Mr. B was given 1 lottery ticket. He won $750 in the lottery.*
- *Who was happier?* Mr. A



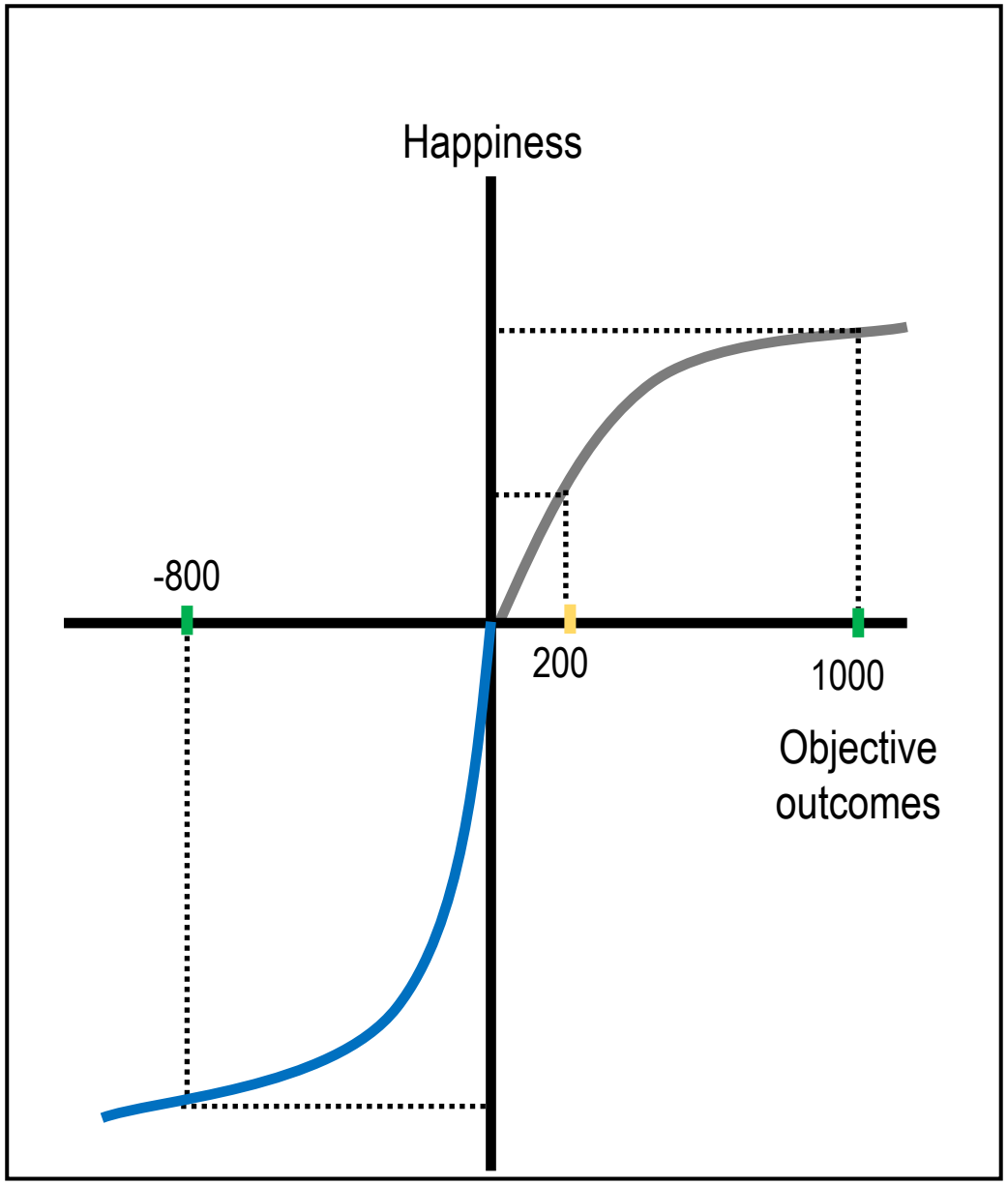
Case 2:

- *Mr. A received a letter from CUHK library saying that he owed them $1000 for losing a book. On the same day, he also received a letter from HKU library saying that he owed them $500 for losing a DVD.*
- *Mr. B received a letter from the University Libraries System saying that he owed them $1500 for losing a book and a DVD.*
- *Who was happier?* _____



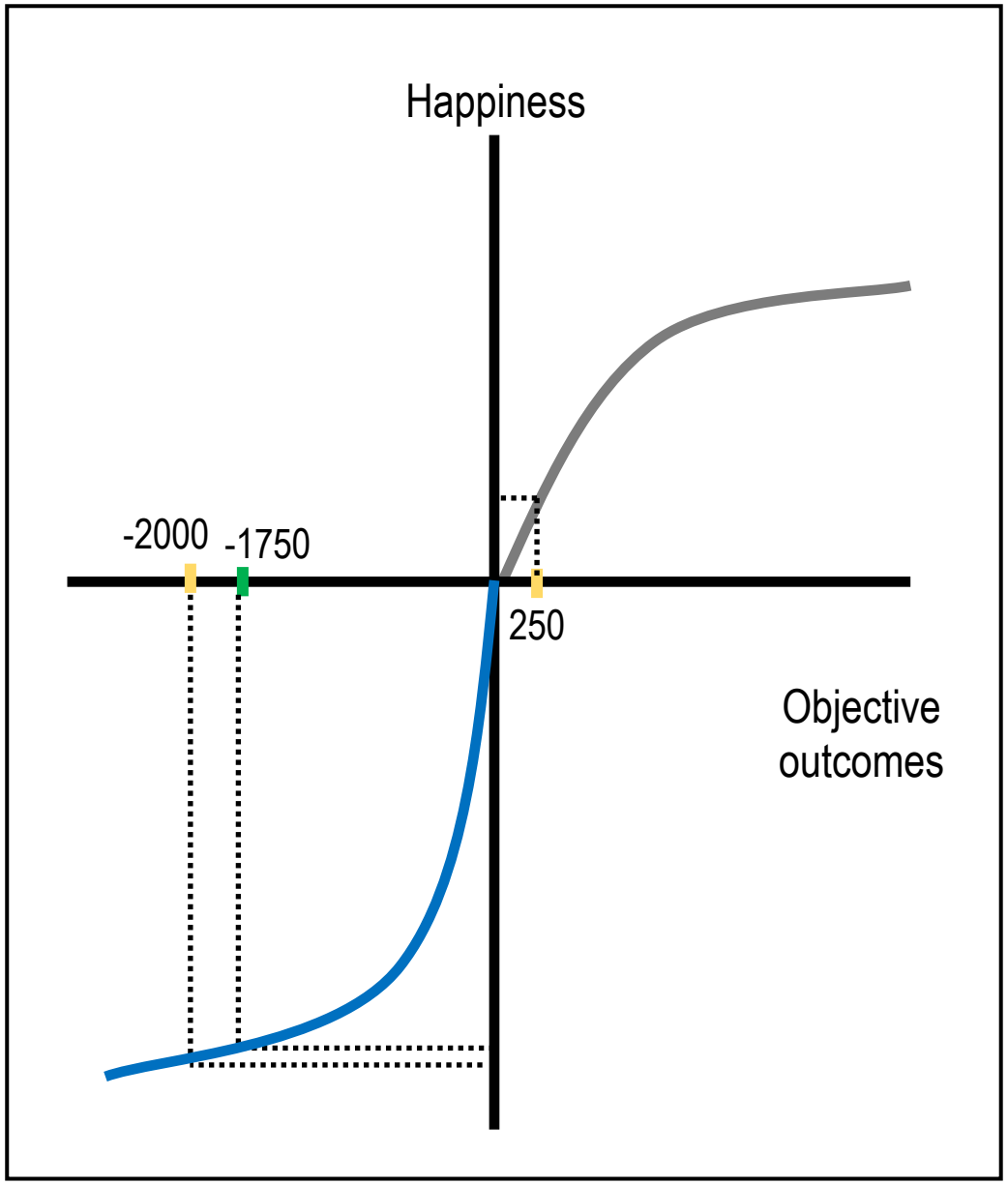
Case 3:

- *Mr. A bought a lottery ticket from a shop and won $1000. While collecting the winnings, he damaged the carpet in the shop and had to pay $800 to repair the carpet.*
- *Mr. B bought a lottery ticket from a shop and got $200 from the shop.*
- *Who was happier?* _____



Case 4:

- *Mr. A went to an academic conference and spent $2000 for the trip. He got $250 from his school as a subsidy.*
- *Mr. B went to a school-subsidized conference trip, and had a net spending of $1750.*
- *Who was happier?* _____



*The above 4 cases illustrate the **principles of hedonic framing***.

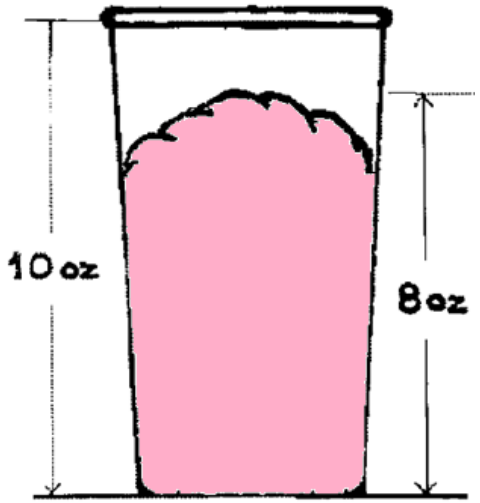
To maximize happiness, we…

- Segregate gains (because of diminishing sensitivity of the value function in the domain of gains).
- Integrate losses (because of diminishing sensitivity of the value function in the domain of losses).
- Integrate smaller losses with larger gains (to offset loss aversion).
- Segregate small gains (sliver linings) from larger losses (because the gain function is steepest at the origin, the utility of a small gain can exceed the utility of slightly reducing a large loss).

❖ *The prospect theory value function can explain preference reversals.*

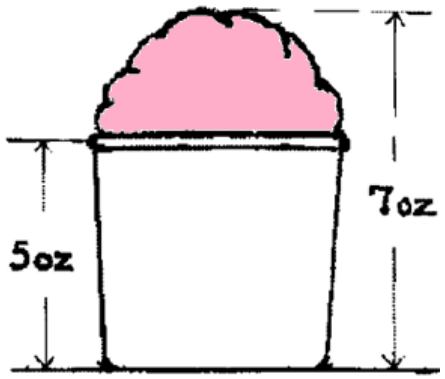
Consider a study with 3 experimental conditions:

Condition 1:
What is the maximum $$ you are willing to pay for a serving?



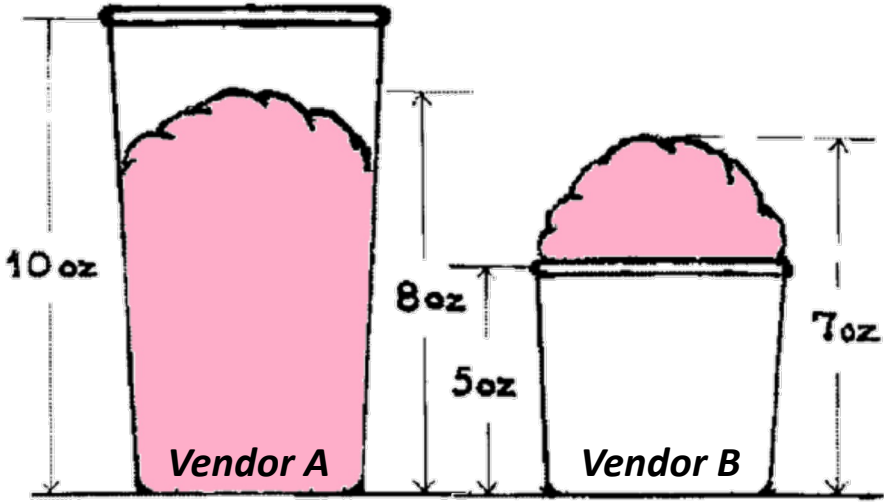
$1.66

Condition 2:
What is the maximum $$ you are willing to pay for a serving?



$2.26

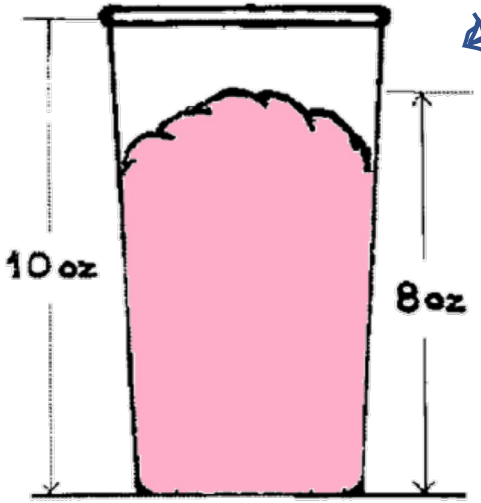
Condition 3:
What is the maximum $$ you are willing to pay for a serving from each of these two vendors?



$1.85 *$1.56*

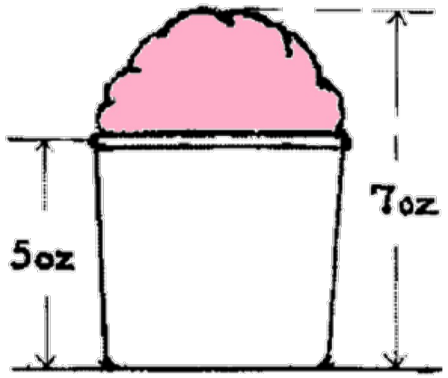
*This is called the **less-is-better effect**: A normatively less valuable option is judged more favorably than a more valuable option.*

- When you are in a *separate evaluation mode* (i.e., the first two experimental conditions), you cannot compare the ice-cream with the other vendor's offering. How do you know how valuable the ice-cream is? The container provides you a *point of reference*.



You don't know how good an 8oz serving is.
But you know that under-filling is no good!

You don't know how good a 7oz serving is.
But over-filling appears to be good!



- When you are in a *joint evaluation mode* (i.e., the 3rd experimental condition), you can compare the offerings from the 2 vendors. Of course, an 8oz serving is better than a 7oz serving!

Hsee, C.K. (1998). Less is better: When low-value options are valued more highly than high-value options. *Journal of Behavioral Decision Making,* 11, 107-121.

Consider another study on preference reversal:

- As part of your professional training you were assigned to a part-time job. The training is now ending and you must look for employment. You consider two options; they are like your training job in most aspects except for (i) the amount of social contact and (ii) the time required to commute to and from work. You would like to have more social contact and less commute time.

Condition 1:

Job	**Contact with others**	**Commute time**
Present job	Isolated for long stretches	10 min.
Job A	Limited contact with others	20 min.
Job B	Moderately sociable	60 min.

Condition 2:

Job	**Contact with others**	**Commute time**
Present job	Most pleasant social interaction	80 min.
Job A	Limited contact with others	20 min.
Job B	Moderately sociable	60 min.

- Results:

Condition 1:

70% of respondents chose Job A

Job	Contact with others	Commute time
Present job	Isolated for long stretches	10 min.
Job A	Limited contact with others	20 min.
Job B	Moderately sociable	60 min.

Condition 2:

33% of respondents chose Job A

Job	Contact with others	Commute time
Present job	Most pleasant social interaction	80 min.
Job A	Limited contact with others	20 min.
Job B	Moderately sociable	60 min.

The options (Jobs A and B) are evaluated against a reference point – the present job. E.g., In Condition 1, the options are *better* than the reference point on *social contact* and *worse* on *commute time*.

Because of loss aversion, an attribute will have a larger impact when in "the domain of losses" than when in "the domain of gains."

Tversky, A., & Kahneman, D. (1991). Loss aversion in riskless choice: A reference-dependent model. *Quarterly Journal of Economics*, 106, 1039-1061.

❖ *The endowment effect*

Consider an experiment:

The seller condition:

- "You now own the mug, you can keep and take it home. You also have the option of selling it and receive money for it."
- A series of prices ranging from $0.25 to $9.25 was listed. "For each of the prices listed below, please indicate whether you wish to (1) sell your mug and receive that amount of money, or (2) keep your mug. After you have finished, one of the prices listed will be selected at random and any exchanges will take place at that price."

The chooser condition:

- For each of the prices, participants were asked to indicate whether they wish to receive (1) a mug, or (2) cash.

Kahneman, D., Knetsch, J.L., & Thaler, R.H. (1990). Experimental tests of the endowment effect and the Coase theorem. *Journal of Political Economy*, 98(6), 1325-1348. Experiment 6.

Results:

- On average, *sellers* asked for at least *$7.12* to sell the mug.
- Whereas, *choosers* decided to receive money rather than the mug at *$3.12*.

Why?

For sellers, their reference point is “having the mug”, they are facing the prospect of *losing* the mug.

For choosers, their reference point is “no mug”, they are facing the prospect of *gaining* the mug.

Due to *loss aversion*, sellers need a larger amount of $$ to compensate the loss of the mug, compared to choosers’ valuation of the mug.

*This is called the **endowment effect**: People place a greater value on something they already possess than on something equivalent that they do not possess. (We value things more when we own them.)*

Implications of endowment effect

Is money-back guarantee good for us?

We fail to notice how our perspective will shift once we have the product at home – we will view returning the product as a loss!

CEOs tend to ask for an unrealistically high price when selling businesses.

They tend to view the divestment as a loss, fail to view the divestment as a healthy renewal of corporate portfolio.

❖ ***The prospect theory value function can explain taxi drivers' (irrational) choice of working hours.***

- In many cities, taxi drivers typically rent their cars for a 12-hour period for a fixed fee. They are then entitled to keep all the revenues they earn during that half-day. Since 12 hours is a long time to drive a car, the drivers must decide each day how long to drive.

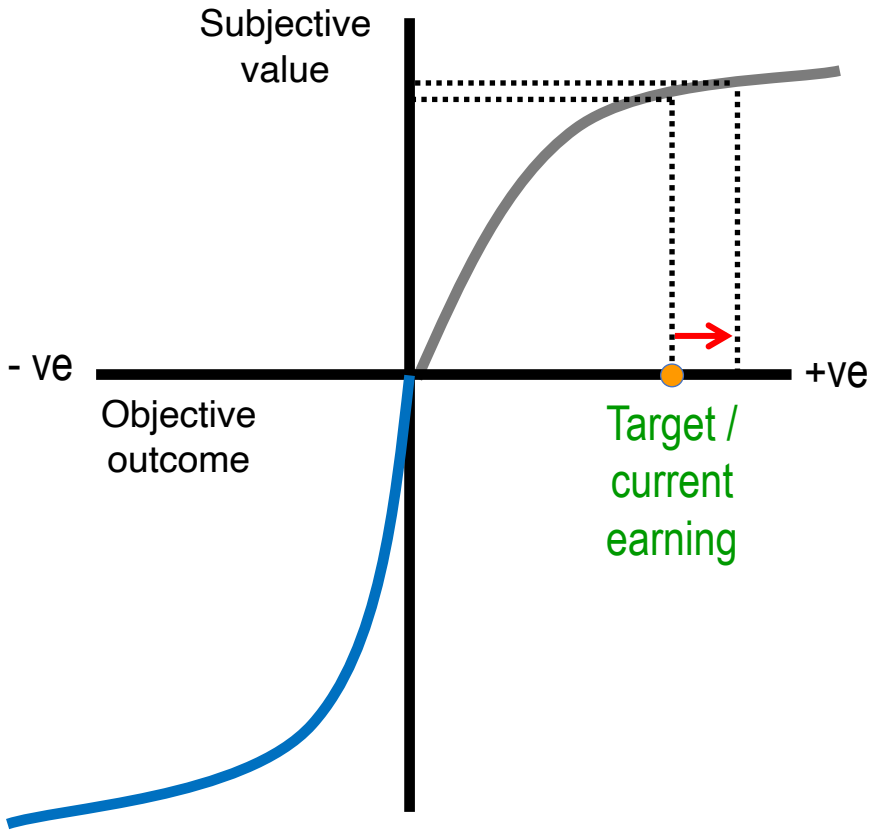
- There is higher demand for taxi service on some days than other days (because of differences in weather, the presence of a big convention, etc.)
- A rational analysis would lead drivers to work long hours on the "fast days" (i.e., the days that demand for taxi service is high), as this practice would maximize earnings per hour worked.

Camerer, C., Babcock, L., Loewenstein, G., & Thaler, R.H. (1997). Labor supply of New York city cabdrivers: One day at a time. *Quarterly Journal of Economics,* 112(2), 407-441.

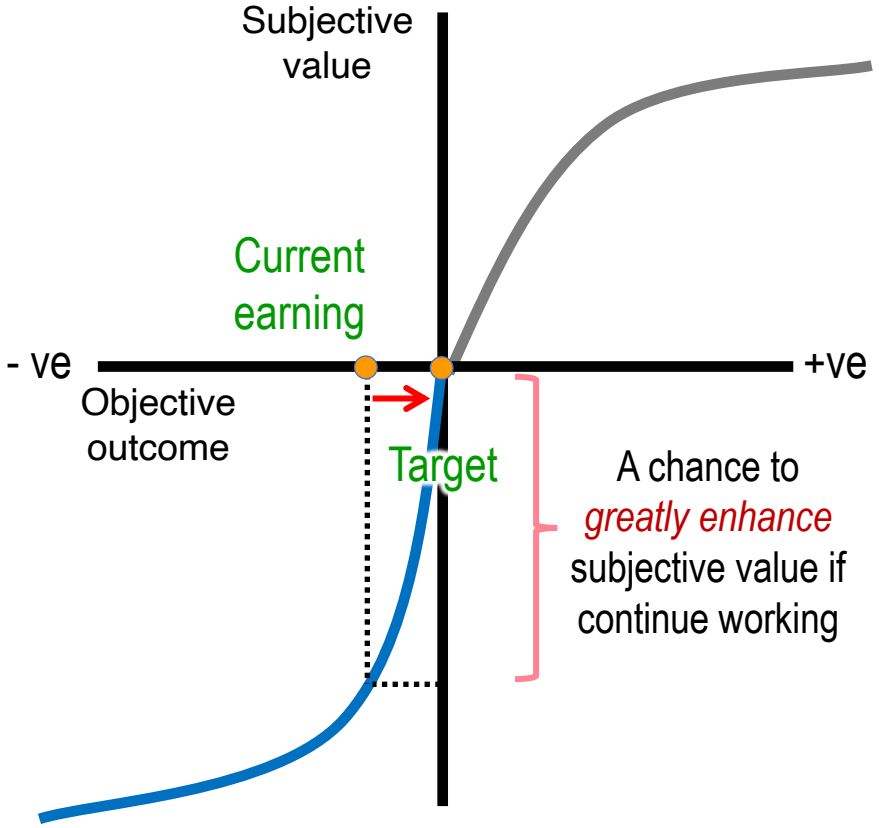
- Drivers, however, tend to work long hours on the “slow days”. Why?

Drivers establish a daily earning target, e.g., HK$1000 per day.

On “fast days”, after reaching their targets, the gain of further earnings does not produce the desire to continue working.



On “slow days”, motivated by the loss of not reaching their targets, drivers tend to work for more hours.



❖ ***The prospect theory value function can explain the effect of choice bracketing on variety-seeking.***

Consider a field experiment conducted on a Halloween night:

- Trick-or-treaters (aged 3-14) approached two adjacent houses.

  The separated choice condition: A choice between two candy bars (3Musketeers and MilkyWay) at each of the 2 houses.

  The combined choice condition: Participants were told at the first house they reached to "choose whichever two candy bars you like."
- Would *choice bracketing* influence the *variety* of candies chosen?

Read, D., & Loewenstein, G. (1995). Diversification bias: Explaining the discrepancy in variety seeking between combined and separated choices. *Journal of Experimental Psychology: Applied*, 1(1), 34-49. Experiment 4.

Results:

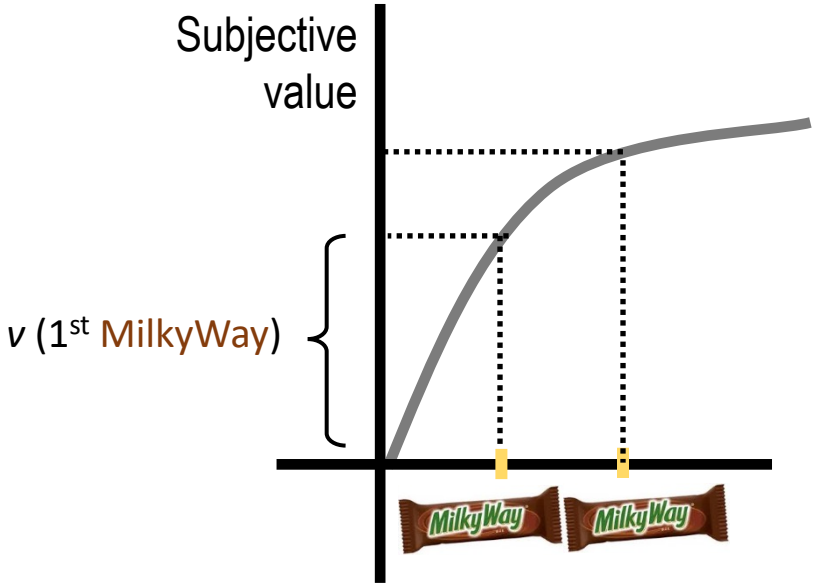
- In the *separated choice* condition, 46% of participants picked one of each candy (i.e., a 3Musketeers and a MilkyWay), 54% picked 2 candies of the same kind (i.e., either 2 pieces of 3Musketeers or 2 pieces of MilkyWay).
- In the *combined choice* condition, 100% of participants picked one of each candy (i.e., a 3Musketeers and a MilkyWay).
- Participants seek more variety in the combined choice condition. Why?

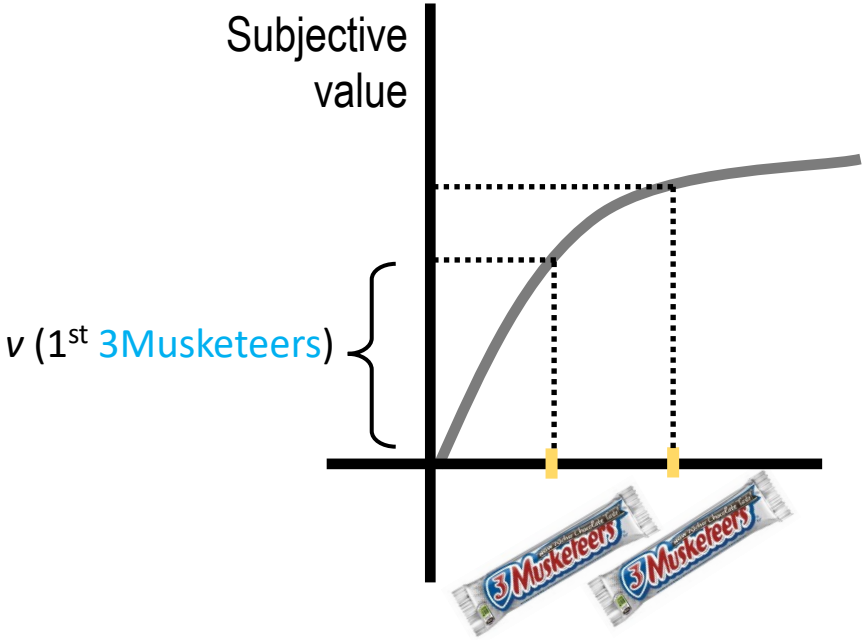
Let's consider a kid who likes MilkyWay more than 3Musketeers, the following graphs show the value functions of the kid for the two types of candy.

- In the *separated choice* condition, the kid made two identical decisions twice (one decision at each house).

  The utility from the 1st piece of MilkyWay is higher than the utility from the 1st piece of 3Musketeers.

  The kid will pick a MilkyWay at the 1st house, and a MilkyWay again at the 2nd house.

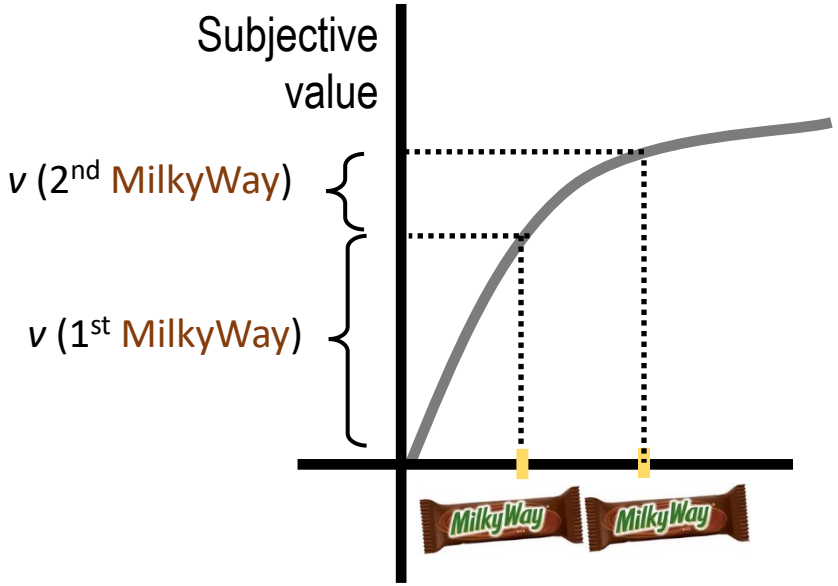


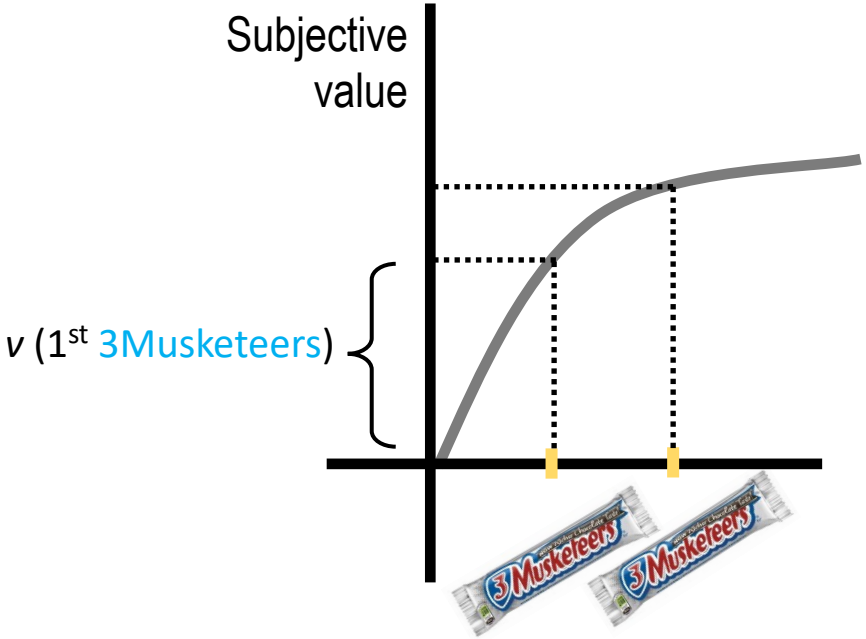


- In the *combined choice* condition, the choices for 2 candies are made together (at the same house).

  The utility from the 1st piece of MilkyWay is higher than the utility from the 1st piece of 3Musketeers, thus the kid will pick a MilkyWay first.

  Then, utility from the 2nd piece of MilkyWay is lower than the utility from the 1st piece of 3Musketeers, thus the kid will pick a 3Musketeers.





Implications of choice bracketing and variety seeking

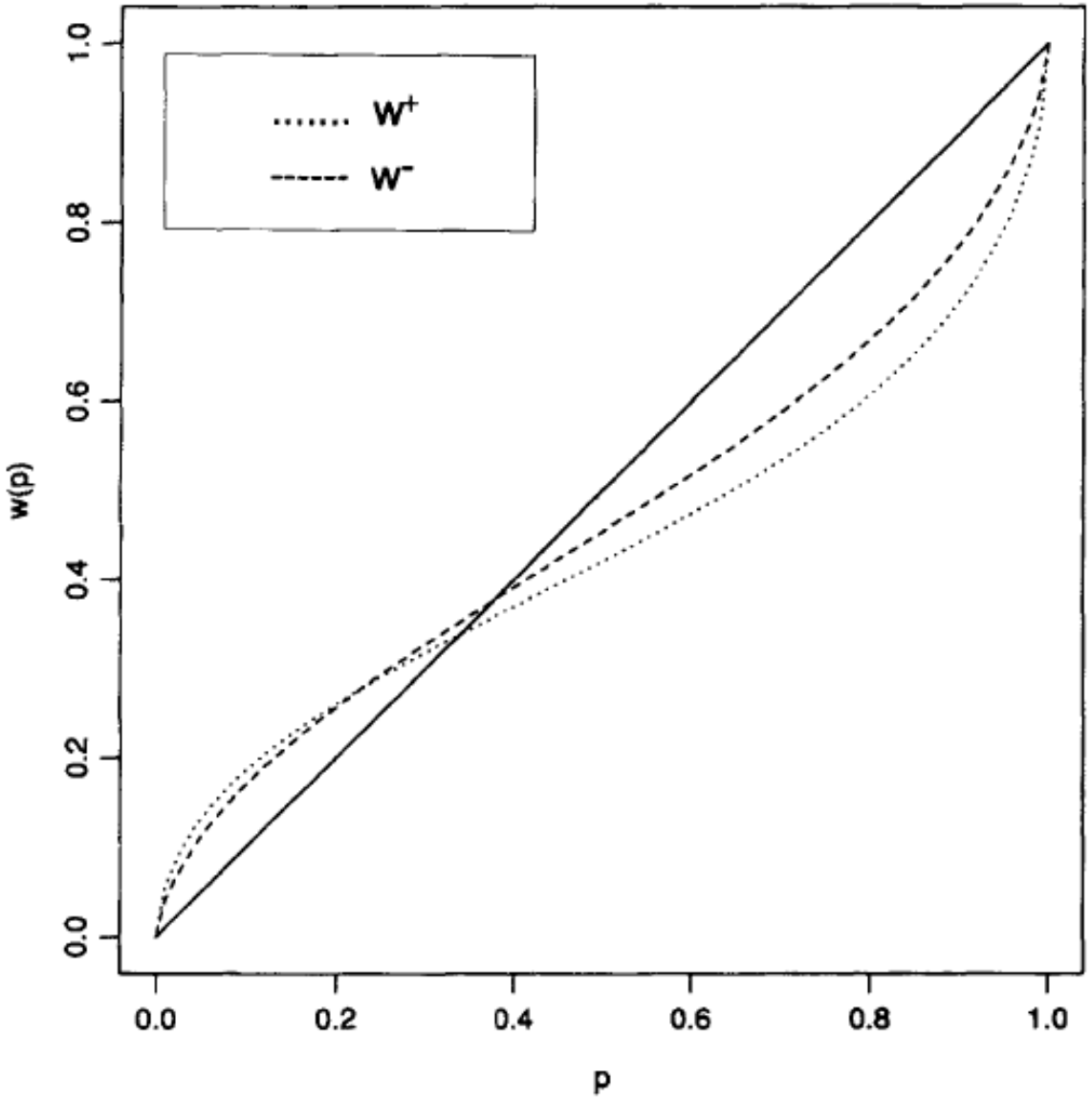
Consumers seek more variety when their choices are bracketed together (i.e., buying in bulk) than when they are made separately.

The unwanted degree of variety when buying in bulk lowers the satisfaction in subsequent consumption occasions.

In some situations, more variety is to our advantage.

For example, nutritionists who wish to facilitate a varied diet could encourage people to simultaneously plan many meals.

PROSPECT THEORY WEIGHTING FUNCTION



Prospect theory's weighting function relates an *objective probability* to people's subjective understanding of the probability (called *decision weight*).

The weighting function has 2 properties:

1) Possibility effect: A change in probability from 0% to a small probability can create a huge psychological impact.
   - When p is very small, w(p) > p.
2) Certainty effect: A change in probability from a large probability to 100% can create a huge psychological impact.
   - When p is very large, w(p) < p.

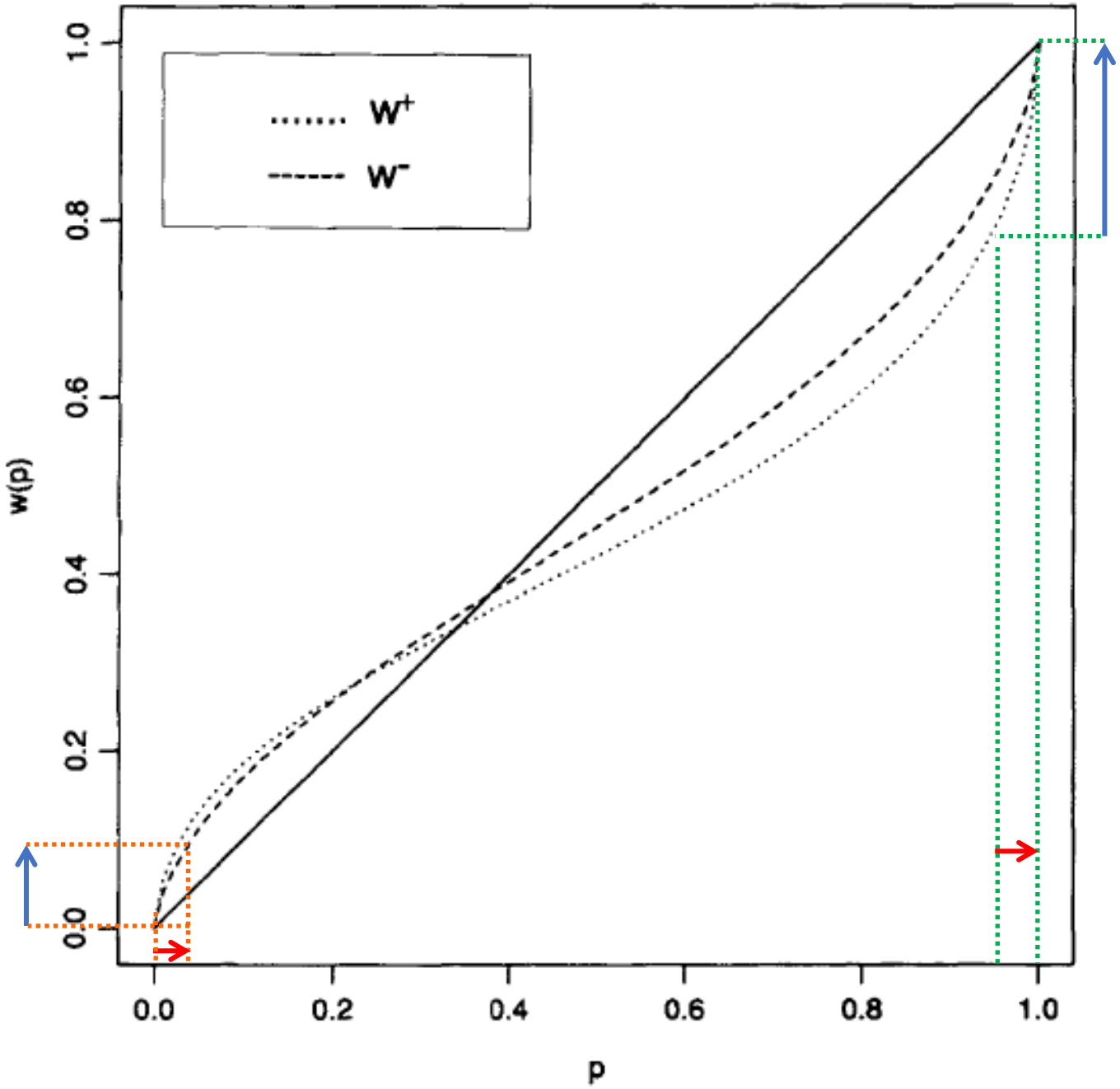
Possibility Effect and Certainty Effect

Take a look on the following 4 changes: In each case, your chance of receiving $1 million improves by 5%. Is the news equally good in each case?

0% → 5%	15% → 20%
60% → 65%	95% → 100%

- The change "0% → 5%" has a large psychological impact, because it creates a possibility that did not exist earlier (*possibility effect*).
- The change "95% → 100%" has a large psychological impact, because it eliminates an uncertainty (*certainty effect*).
- Certainty and possibility have similarly strong effects in the domains of gains and losses.

Small change in probability to create a certainty → large change in decision weight



Small change in probability to create a possibility → large change in decision weight

USING PROSPECT THEORY TO PREDICT RISK PREFERENCE

Value function

+

Weighting function

→

Prospect theory enables us to predict people's risk preferences (i.e., tendency to accept a sure option vs. a risky option when the EMVs of the 2 options are the same).

$$\mathbf{EU(option) = \Sigma\, v(x_n)\, w(p_n)}$$

We can compare EU(sure option) with EU(risky option) to see which one is higher.

Consider the following experiment:

Imagine that the U.S. is preparing for the outbreak of an unusual Asian disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimates of the consequences of the programs are as follows, which of the two programs would you favor? Program A or B?

Condition 1 [***Gain frame***]

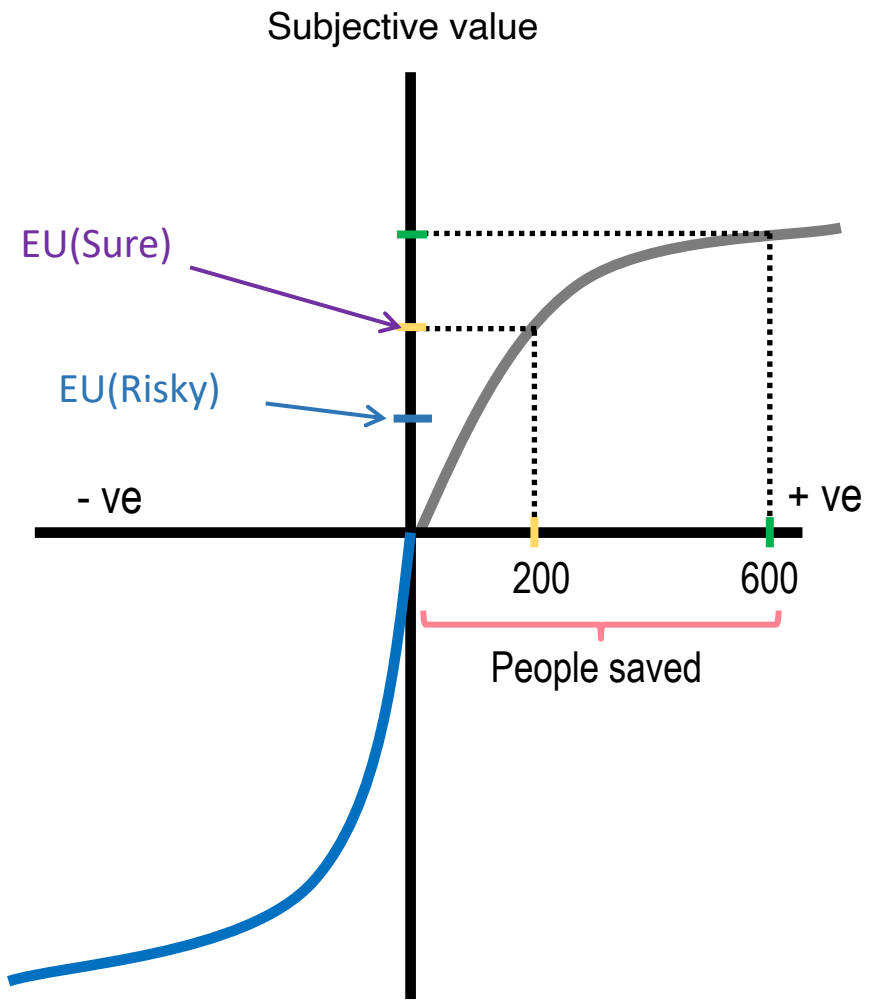
- If Program A is adopted, 200 people will be saved. [sure option] 72%
- If Program B is adopted, there is 1/3 probability that 600 people will be saved, and 2/3 probability that no people will be saved. [risky option] 28%

Condition 2 [***Loss frame***]

- If Program A is adopted, 400 people will die. [sure option] 22%
- If Program B is adopted, there is 1/3 probability that nobody will die, and, and 2/3 probability that 600 people will die. [risky option] 78%

Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, 211, 453-458.

Condition 1 [Gain Frame]



- If Program A is adopted, 200 people will be saved.
- If Program B is adopted, there is 1/3 probability that 600 people will be saved, and 2/3 probability that no people will be saved.

$$\boxed{EU(option) = \Sigma\, v(x_n)\, w(p_n)}$$

What is the EU of Program A?
$v(200)\, w(1)$

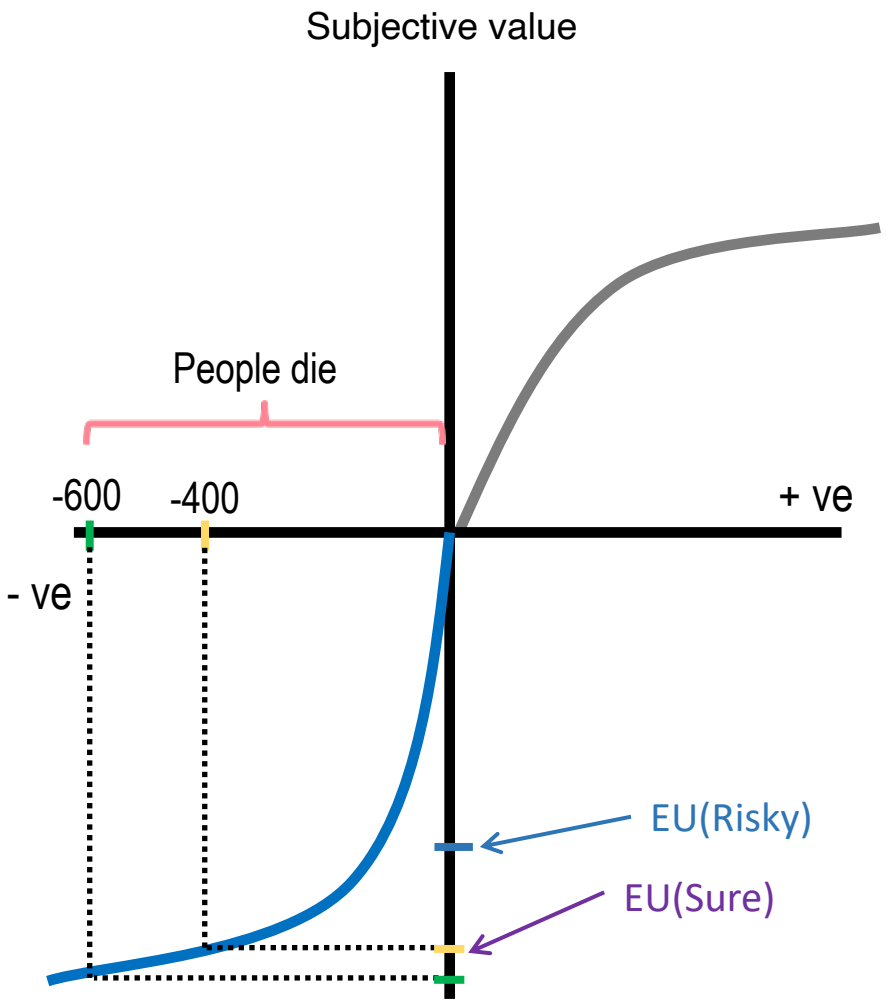
What is the EU of Program B?
$v(600)\, w(1/3) + v(0)\, w(2/3)$

Reaction to saving 200 people is stronger than 1/3*reaction to saving 600 people.

The decision weight that corresponds to a probability of 1/3 is slightly higher than 1/3.

Thus, people prefer the sure option.

Condition 2 [Loss Frame]



- If Program A is adopted, 400 people will die.
- If Program B is adopted, there is 1/3 probability that nobody will die, and 2/3 probability that 600 people will die.

$$EU(option) = \Sigma\, v(x_n)\, w(p_n)$$

What is the EU of Program A?
$v(-400)\, w(1)$

What is the EU of Program B?
$v(0)\, w(1/3) + v(-600)\, w(2/3)$

Reaction to a loss of 400 people is stronger than 2/3*reaction to a loss of 600 people.

The decision weight that corresponds to a probability of 2/3 is slightly less than 2/3.

Thus, people prefer the risky option.

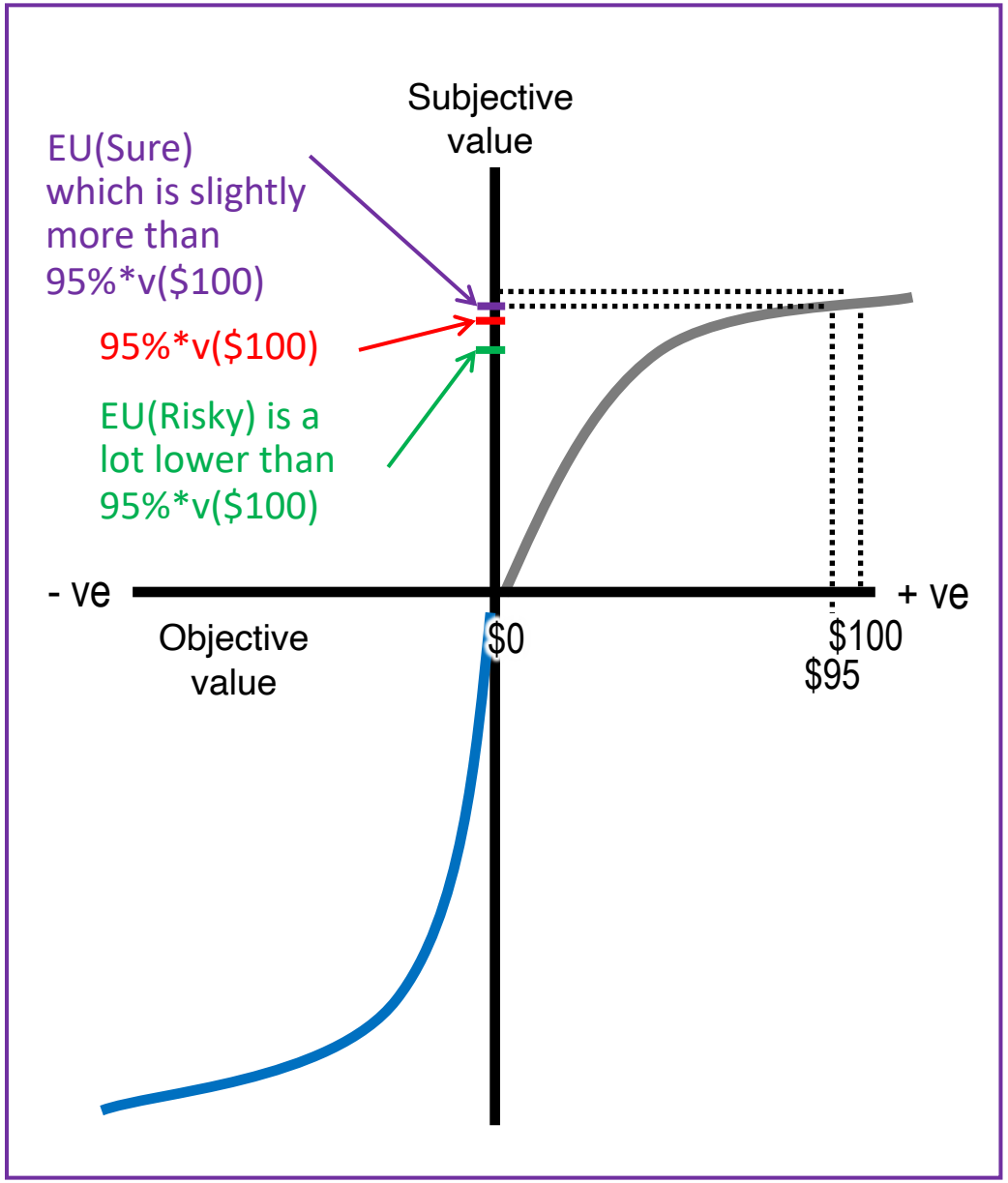
Predictions of Prospect Theory

	Gains	Losses
High probability	95% chance to win $100 vs. sure win of $95 **RISK AVERSE**	95% chance to lose $100 vs. sure loss of $95 **RISK SEEKING**
Low probability	5% chance to win $100 vs. sure win of $5 **RISK SEEKING**	5% chance to lose $100 vs. sure loss of $5 **RISK AVERSE**

- The first row in each cell shows an illustrative prospect.
- The second row indicates how most people behave when offered a choice between a gamble and a sure gain (or loss) that corresponds to the gamble's expected value. Choices are said to be risk averse if the sure thing is preferred, risk seeking if the gamble is preferred.

[The upcoming 4 slides are illustrations of the 4 examples mentioned above.]

Supplementary Information



Domain = Gains, Probability = Large

- Risky option: 95% chance to win $100
- Sure option: 100% chance to win $95

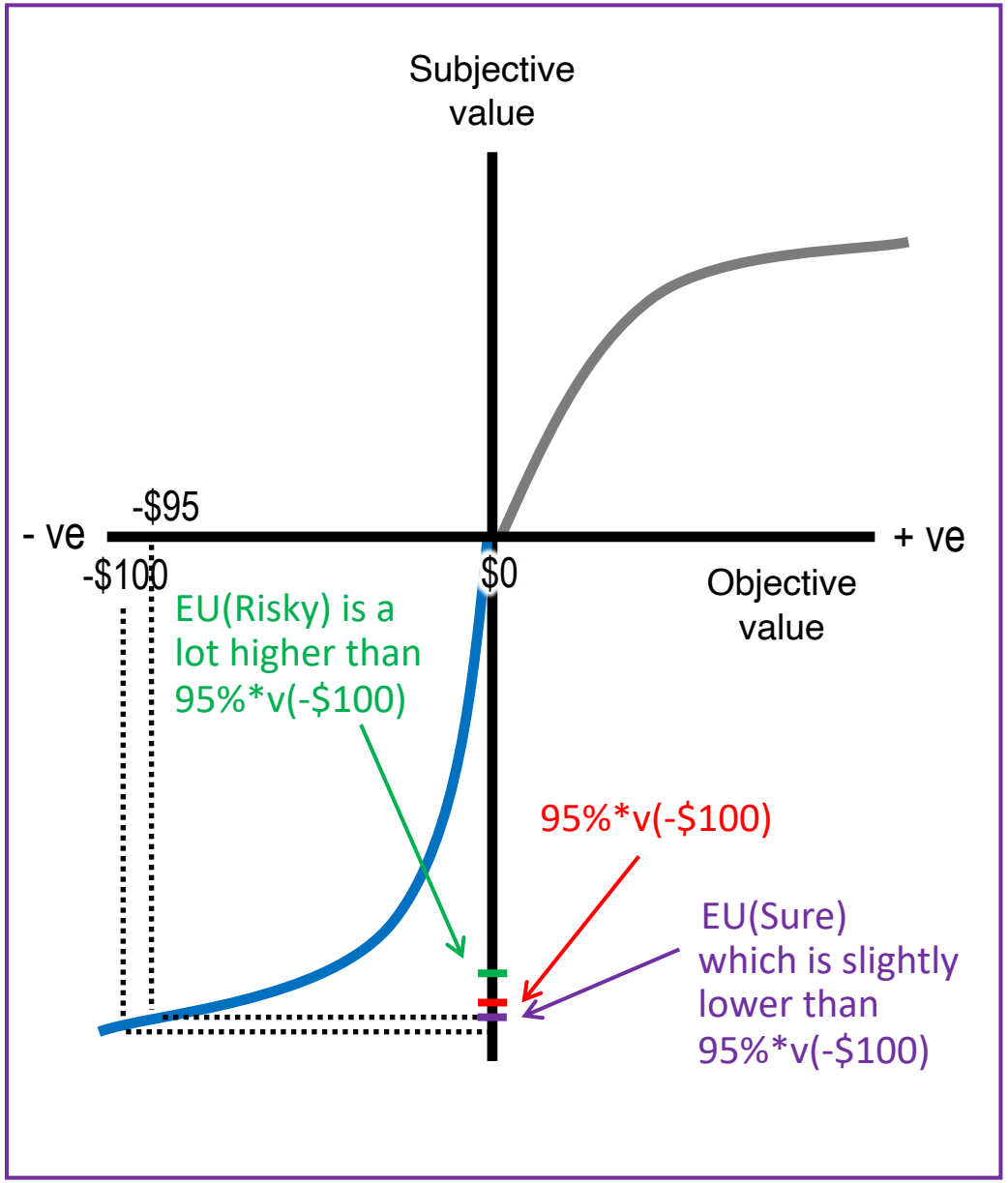
What is the EU of the sure option?
v($*95*) w(*1*)

What is the EU of the risky option?
v($100) w(*0.95*) + *v($0*) w(*0.05*)

Reaction to a gain of $95 is slightly stronger than 95%*reaction to a gain of $100.

The decision weight that corresponds to a probability of 0.95 is a lot smaller than 0.95.

Thus, people prefer the sure option to avoid potential disappointment caused by the risky option.



Supplementary Information

Domain = Losses, Probability = Large

- Risky option: 95% chance to lose $100
- Sure option: 100% chance to lose $95

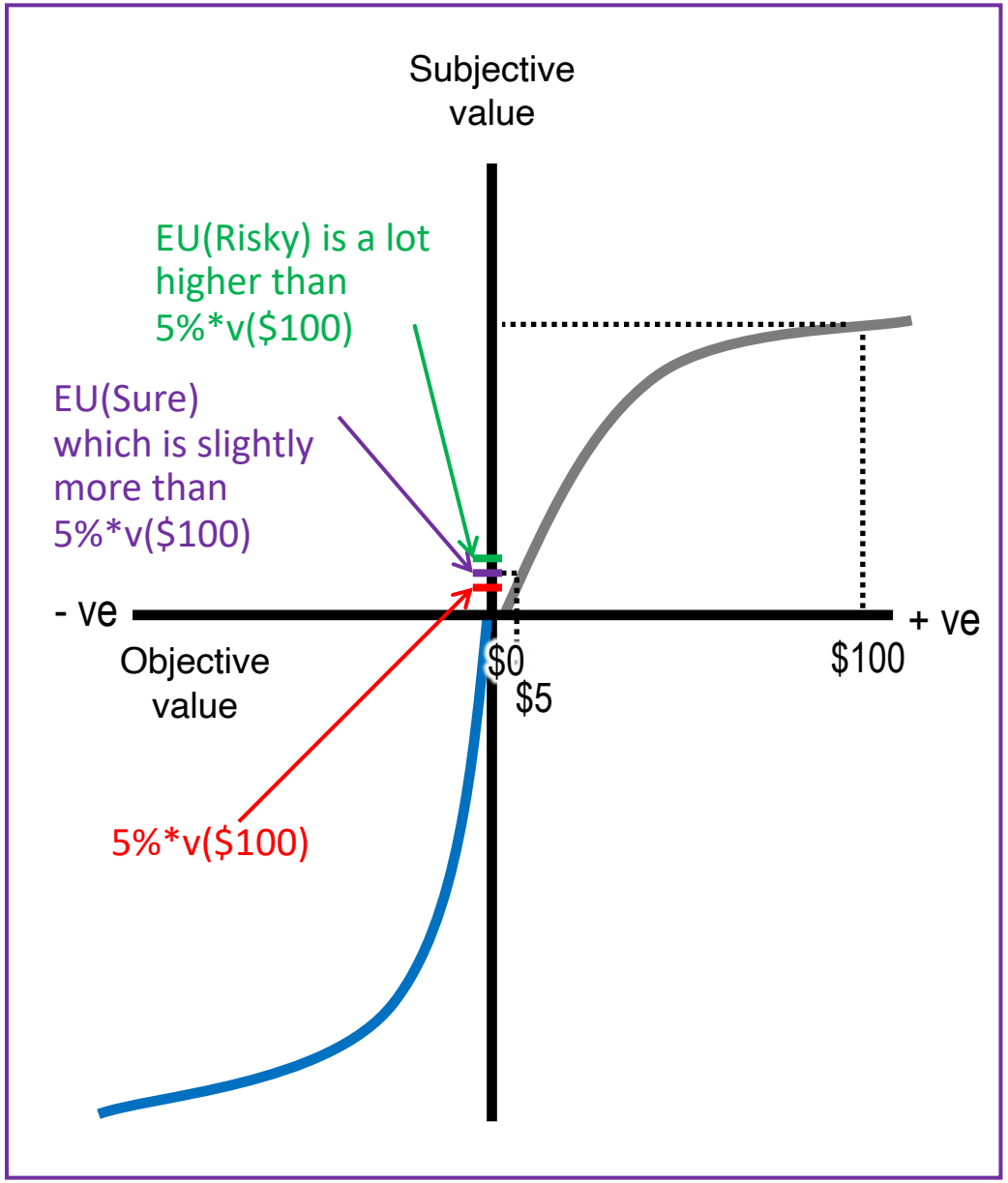
What is the EU of the sure option?
v(-$*95*) w(*1*)

What is the EU of the risky option?
v(-$100) w(*0.95*) + *v($0)* w(*0.05*)

Reaction to a loss of $95 is slightly stronger than 95%*reaction to a loss of $100.

The decision weight that corresponds to a probability of 0.95 is a lot smaller than 0.95.

Thus, people prefer the risky option, hope to avoid loss.



Supplementary Information

Domain = Gains, Probability = Small

- Risky option: 5% chance to win $100
- Sure option: 100% chance to win $5

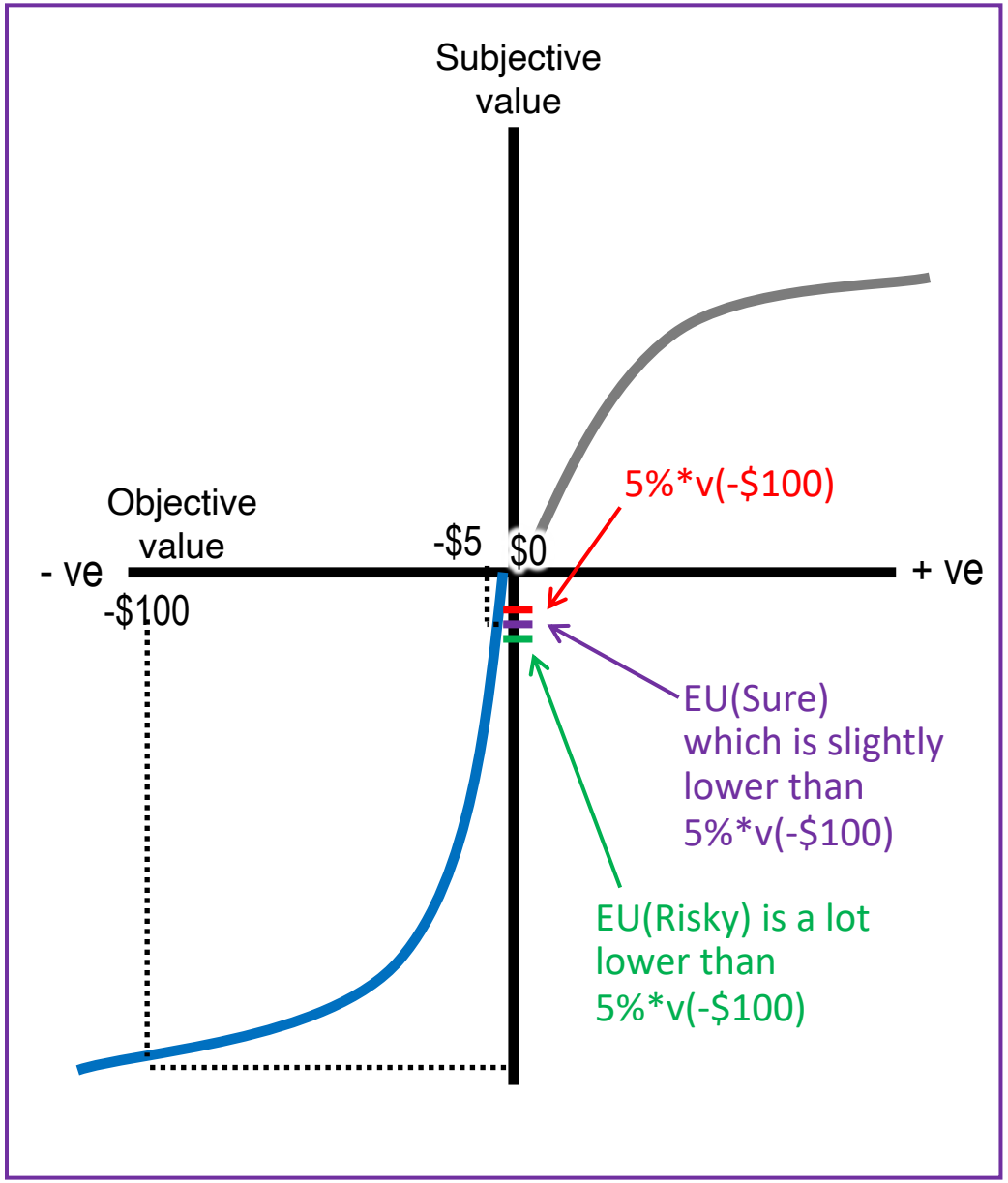
What is the EU of the sure option?
v($5) w(1)

What is the EU of the risky option?
v($100) w(*0.05*) + *v($0)* w(*0.95*)

Reaction to a gain of $5 is slightly stronger than 5%*reaction to a gain of $100.

The decision weight that corresponds to a probability of 0.05 is a lot larger than 0.05.

Thus, people prefer the risky option. This explains why lotteries are popular.



Supplementary Information

Domain = Losses, Probability = Small

- Risky option: 5% chance to lose $100
- Sure option: 100% chance to lose $5

What is the EU of the sure option?
v(-$*5*) w(*1*)

What is the EU of the risky option?
v(-$100) w(*0.05*) + *v($0)* w(*0.95*)

Reaction to a loss of $5 is slightly stronger than 5%*reaction to a loss of $100.

The decision weight that corresponds to a probability of 0.05 is a lot larger than 0.05.

Thus, people prefer the sure option. This explains why people buy insurance.

Intended Learning Outcomes For Unit 2.5

By now, you should be able to:

- Describe the characteristics of prospect theory's value function (i.e., reference dependence, diminishing sensitivity, and loss aversion), and describe its resulting phenomena (e.g., principles of hedonic framing, endowment effect).
- Describe the characteristics of prospect theory's weighting function (i.e., certainty effect and possibility effect).
- Predict people's risk preferences under various combinations of frame of outcome (gain, loss) and probability (low, high) using the prospect theory.

Time For Reflection

Which idea(s) in this class surprised you most?
What's your key takeaway?